

Thermal degradation of municipal plastic waste for production of fuel-like hydrocarbons

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Abstract

Fuel-like utilization is one way of chemical recycling of liquids from waste polymers. The degradation of different mixtures of the most frequently occurring waste polymers (polyethylene, polypropylene, polystyrene, polyamide, ethylene-propylene copolymer and polyurethane rubber) was investigated in a horizontal tube reactor. The effects of cracking temperature and residence time on the yields and structure of the products were studied. The gas and liquid products formed were analysed by gas chromatography. In addition, the chemical structure of waste polymer had a considerable effect on the properties of the products. With increasing temperature, the carbon atom distribution of the hydrocarbons obtained became wider and the residence time had a significant effect upon the structure of the products. The liquid products had low heteroatom content (sulphur, nitrogen), but 2% of polyurethane rubber and 2% of polyamide in the raw material polyamide increased the concentration of sulphur and nitrogen in the products. © 2004 Elsevier Ltd. All rights reserved.

Keywords: Municipal plastic waste; Tube reactor; Thermal degradation; Residence time; Gas chromatography; Aromatic content; Sulphur and nitrogen content

1. Introduction

The growing amount of plastic waste is generating more and more environmental problems worldwide. The present rate of economic growth is unimaginable without saving of fossil energy like crude oil, natural gas or coal. Suitable waste management is another important aspect of sustainable development. Plastic wastes represent a considerable part of municipal wastes, furthermore huge amounts of plastic waste arise as a by-product or faulty product in industry and agriculture. According to estimates, plastic wastes represent 15–25% of municipal waste. The amount of plastic materials was 25 Mt in Europe and it will reach 35 Mt by 2010 [1,2].

Nowadays there are three ways to utilize plastic waste: landfilling, incineration—with or without energy recovery—and recycling. The largest amount of plastic

wastes is disposed of by landfilling (65–70%), and incineration (20–25%). Recycling is only about 10%. Moreover, the problem of wastes cannot be solved by landfilling and incineration, because suitable and safe depots are expensive, and incineration stimulates the growing emission of harmful, greenhouse gases, e.g. NO_x, SO_x, CO_x, etc. Recycling can be divided into further important categories, such as mechanical recycling and chemical recycling. Chemical recycling is virtually a thermal method by which the long alkyl chains of polymers are broken into a mixture of lighter hydrocarbons. This is one of the prospective ways to utilize waste polymers [3–9].

Thermal recycling of waste polymers under different catalytic and thermal circumstances has been well investigated by researchers [10–15]. It was found that both the yields and chemical properties of products can be modified with catalysts. In batch reactions, catalysts can be added easily, but in a continuous system it might be problematical, because, e.g. the maintenance of fluid

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beds depends on the changing properties of wastes. A further problem is the handling of the deactivated catalysts, which have to be separated from the residue and reactivated. Hardman et al. [16] used silicates and sands for creating a fluidized bed. The minimum temperature of cracking could be reduced to 450–550 °C and later to 430 °C. Sharrath et al. [17] investigated the catalytic degradation of HDPE on a fluidized bed using HZSM-5 catalyst. Higher yields of gases and liquids and higher concentrations of branched hydrocarbons were found with increasing temperature and the presence of catalyst.

Usually batch experiments are carried out at laboratory scale (1–500 g) using one or two types of polymers, which are clean and not waste polymers. On the other hand, the effects of the cracking temperature on the properties of products are well demonstrated in the literature not only in batch processes but also in continuous or semi-continuous cracking. Murata et al. [18] investigated the cracking of PE, PP and PS in a continuous flow stirred tank reactor. It was found that the degradation by continuous flow operation is a suitable technique for converting waste polymers into liquid hydrocarbon. The volatile products could be used as feedstock components, e.g. in refineries. Their further utilization for petrochemical purposes has not yet been solved. One possibility is a fuel-like application or mixing in fuels as a blending component. Before blending the high olefin content is to be saturated with hydrogen, or hydroisomerized. These steps result in a high quality synthetic diesel fuel, with high cetane number, and theoretically free from sulphur, nitrogen and metals. In practice these fractions generally have a very low heteroatom content, even if the raw material was pure and not waste, because most polymers contain, e.g. sulphur containing anti-flame or antioxidant additives, etc.

Other important parameters for waste polymer degradation are the residence time, and the chemical structure of polymers. The effects of these parameters have been less investigated and described. The packaging industry and other human activities use a huge amount of PE and PP. In a company collecting and handling communal and selective wastes, separation cannot be ideal thus the targeted polymer type always contains also other types of polymers. The processed materials are therefore polymer mixtures.

The subject of this paper is the investigation of the cracking behaviour of municipal plastic waste. The effects of cracking parameters (e.g. residence time, temperature) on the properties of the product were investigated in a horizontal tube reactor, over a range of temperature ($T = 490$ – 550 °C) and residence time ($\tau = 0.6$ – 1.2 h). Both yields and volatile product properties, e.g. olefin content, carbon number distribution, heteroatom content, etc. were investigated.

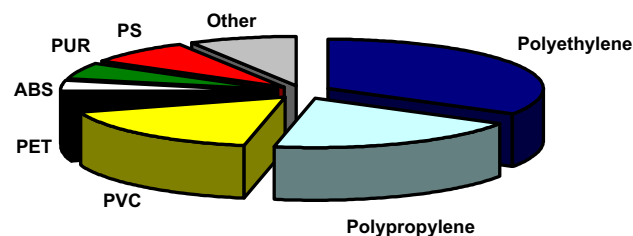


Fig. 1. The composition of municipal plastic waste.

2. Experimental

2.1. Raw materials

The average composition of municipal plastic waste (MPW) in Europe is shown in Fig. 1 [1,2]. Based on the composition of this average plastic mixture, a commercial polyethylene (PE), a polypropylene (PP), polystyrene (PS), ethylene-propylene copolymer (EPC), polyamide (PA) and polyurethane rubber (PUR) were used as raw materials in our experiments. Each polymer was obtained as wastes of industrial or agricultural processes. The polyethylene waste was chosen from an agricultural foil, the polypropylene and polystyrene were selected from packaging wastes. The ethylene-propylene copolymer waste originated from parts of computers and the polyamide and polyurethane rubber were obtained from by-products of the car manufacturing industry. The main properties of these waste polymers are summarized in Table 1.

Two mixtures of the polymers given in Table 1 were used as MPW-1 and MPW-2. MPW-1 had 40% PE, 40% PP and 20% ethylene-propylene copolymer content, while MPW-2 consisted of 39% PE, 49% PP, 2% PA, 2% PUR and 8% PS. One of the MPWs (MPW-2) also had heteroatom content: sulphur from polyurethane rubber and nitrogen from polyamide and polyurethane rubber.

Table 1
The main properties of waste polymers

Properties	PE	PP	EPC	PS	PA	PUR
Grain size (mm)	4–6	4–5	<6	3–4	2–3	3–4
MFI (g (10 min)^{-1}) (2160N)	0.538 ^a	0.376 ^a	0.439 ^a	0.628 ^b	4.736 ^a	22.92 ^b
Density (g cm^{-3})	0.918	0.952	0.938	1.040	1.129	1.223
Ash content (%)	0.67	1.14	3.01	0.01	2.21	0.02
Humidity (%)	0.13	0.28	0.04	0.02	0.96	0.36
Metals (ppm)						
Ca	206	222	801	0	0	0
Zn	38	261	956	102	561	62
Ti	593	103	917	102	120	101

^a At 230 °C.

^b At 190 °C.

2.2. Cracking apparatus

The decomposition of plastic waste was carried out in a horizontal tube reactor, which is schematically shown in Fig. 2. The cracking equipment consisted of three main parts: an extruder, a reactor and a separator. The waste polymers with suitable grain size were fed in by an extruder, where the polymer was preheated up to 250–280 °C. It was directly connected to the reactor, therefore the preheated polymer was driven from the extruder into the reactor. Virtually the cracking reactor had three parts. Each part had the same geometrical properties (e.g. length, diameter, etc.), and they were kept at the same cracking temperature. Inside the reactor the polymers were melted and their carbon chain cracked into low molecular weight fragments. In a separator the hydrocarbons formed were separated into volatile products and residue. The volatile fraction was further separated in a condenser as gas and liquid. Cooling water at 20 °C was used for condensation. Non-condensable gases were passed through a flow meter and then flared. Gases at set intervals were sampled and analysed by a gas chromatograph. At room temperature, solid residue was collected as a bottom product of a control system fitted with thermometers. The degradation behaviour of polymer mixtures was investigated at three different temperatures (500, 525, 550 °C) and residence times (0.6, 0.9, 1.2 h). Liquids were collected and further analysed.

2.3. Analysis of products

In the present work the change of product properties was studied with different cracking parameters. Both gas and liquid products formed in cracking reactions were analysed by standard and non-standard methods.

Gases were analysed with a Carlo-Erba Vega Series GC 6000 gas chromatograph (GC) provided with a 50 m×0.32 mm fused silica column with Al₂O₃/KCl coating, at 40 °C.

The distribution of carbons in liquids was determined by gas chromatography. For this purpose a TRACE GC gas chromatograph was used with a 30 m×0.32 mm Rtx®-1 (Crossbond® 100% dimethyl-polysiloxane) column. The temperature program for the analysis started at 40 °C, then rose by 15 °C min⁻¹ to 330 °C.

The olefin content of liquids was obtained by a method based on an infrared technique for the determination of the distribution of olefin double bonds. A SHIMADZU IR-470 type spectrometer was used for the infrared analysis in the wavenumber range of 400–4000 cm⁻¹. The vinyl double bonds have two significant IR adsorption bands at 910 and 990 cm⁻¹ while vinylidene has internal double bonds at 890 and 956 cm⁻¹. The number of –CH₂– and –CH₃ groups were determined also by IR technique in the range of

2800–3100 cm⁻¹, i.e., in the C–H stretching vibration range of hydrocarbons. The number of the –CH₂– groups is proportional to the intensity of the asymmetric stretching vibration band at 2927 cm⁻¹ and the number of the –CH₃ groups to that of asymmetric stretching vibration band at 2958 cm⁻¹. In our analysis, other standard methods were used as follows:

- liquid density measurement (MSZ EN ISO 12185),
- determination of distillation curve (ASTM-D 1078),
- determination of iodine-bromine number (ASTM-D 149-60),
- determination of sulphur and nitrogen content of liquids (ASTM D 6428 99).

3. Results and discussion

The effect of the cracking parameters (temperature, residence time, chemical structure of raw materials) on the yields and structure of products was investigated by the properties of three different products formed in the laboratory cracking reactions, gas, liquid, residue, with components of different length of hydrocarbon chains. The yields of each fraction obtained in the cracking process are shown in Table 2.

It was found that the yields of volatile products increased both with residence time and temperature in the case of each mixture of polymers. The growing yield of volatile components (e.g. gases and liquids) could be caused by the differences in the thermal stability of polymer chains. Both the linear and branched hydrocarbons have decreasing thermal stability with increasing temperature. Therefore at 550 °C the C–C bonds cracked more likely than at lower, e.g. 500 °C temperature, and this resulted in higher yields of volatile products. This fact was also visible on the physical appearance of the products. The residue became darker and a bit softer after cracking at higher temperature. The yield increasing effect of the residence time is due likewise to the thermal stability of C–C bonds. The probability of cracking of the carbon chain increased with the residence time of thermal degradation, therefore stricter degradation conditions resulted in lighter hydrocarbons with a shorter carbon chain. At a residence time of 1.2 h, the melted polymer stayed longer under thermal degradation conditions, therefore resulting in higher liquid and gas yield. Furthermore, the chemical structure of raw materials also had an effect on the yields. The yields of volatile products for MPW-2 are higher than those for MPW-1 (Table 2). MPW-2 contained 10% more polypropylene waste than MPW-1. Polypropylene consists of a branched hydrocarbon chain, and the carbon atoms in the ramification had lower thermal stability than the others. Therefore the –[CH₂C(CH₃)CH₂]_n– structure has been more easily

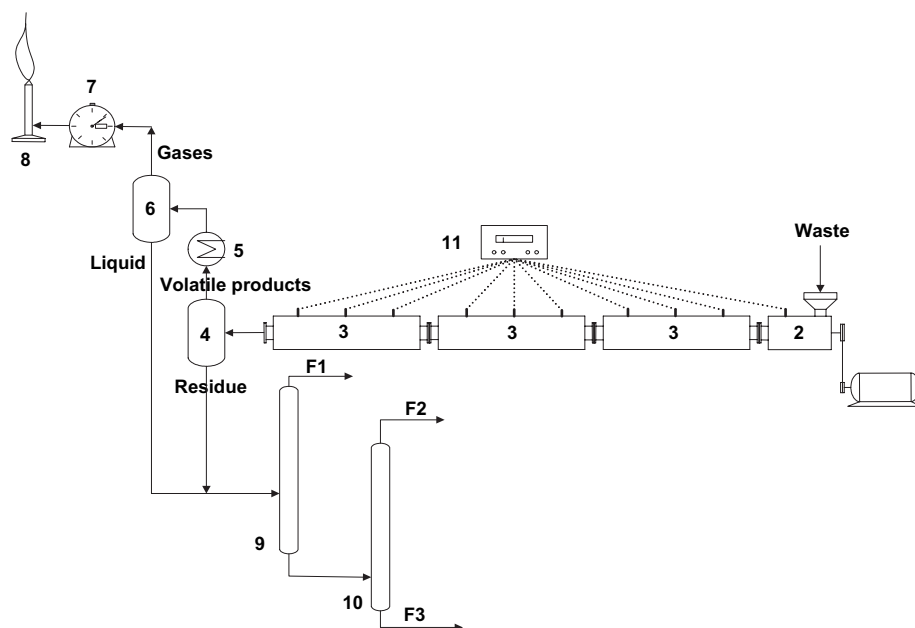


Fig. 2. The process for cracking of municipal plastic waste and separation of the products. 1, motor; 2, extruder; 3, reactor; 4, separator; 5, condenser; 6, separator; 7, gas-flow meter; 8, flare; 9, atmospheric distillation; 10, vacuum distillation; 11, thermometer.

cracked than the $-\text{[CH}_2\text{--CH}_2\text{]}_n-$ structure, and resulted in higher yields of volatile products.

The composition of gases formed in the cracking reactions of polymer blends was analysed with a gas chromatograph at 40 °C. According to our results the cracking parameters (e.g. temperature, residence time) did not affect significantly the composition of gaseous products. On the other hand, significant differences were observed in the case of polymer blends with different composition. Fig. 3 represents the composition of gases formed in cracking reactions at 420–430 °C temperature range and 0.9 h residence time. The composition of gases was analysed in the case of both samples at each temperature and residence time. No significant differences were observed with regard to temperature on residence time.

It was found that in case of both MPW-1 and MPW-2 the gases contained mostly C_2 , C_3 and C_4

hydrocarbons. In the case of MPW-2, with lower PE/PP ratio, the concentration of ethylene and ethane was lower by 10–15% than in case of MPW-1, while the concentration of C_3 fraction was higher by about the same amount. This is a consequence of the composition of the raw materials; sample MPW-1 contains 10% more polyethylene than sample MPW-2. It was found that the concentration of alkanes was higher than that of alkenes. In gas products, the probability hydrogen transfer reactions occurring was greater than that of the β -scission reactions. This agrees with the findings of other researchers, who demonstrated that the cracking of polyolefins resulted in greater concentration of alkanes than alkenes mostly in C_3 and C_4 fractions [19].

The composition of liquid fractions formed in the cracking reactions of municipal plastic wastes was analysed by GC. Identifying individual compounds in liquid products, especially the branched molecule from polypropylene, is rather difficult. It is well known that the cracking of polypropylene yields a great number of isomer compounds. The liquid product obtained by cracking of polyethylene consisted of mostly n-alkanes and n-alkenes, which were evenly distributed by carbon number. The cracking of polystyrene yielded blends of aromatic compounds, styrene, ethyl-benzene, benzene, toluene. Identifying the degradation products of polyamide and polyurethane is difficult, but the formation of R--NH--R' type compounds is quite likely, because the probability of formation of monomer without an adequate catalyst is low.

Table 3 represent the results of gas chromatographic analysis. The two different polymer wastes yielded

Table 2

The yields of cracked products of MPW-1 and MPW-2 raw material mixture

Temperature (°C)	500 °C			525 °C			550 °C		
Residence time (h)	0.6	0.9	1.2	0.6	0.9	1.2	0.6	0.9	1.2
(a) MPW-1									
Gas	1.1	2.3	4.1	1.9	2.7	5.0	3.0	3.4	6.1
Liquid	6.6	12.2	15.4	11.1	13.4	17.2	17.5	19.4	24.3
Residue	92.3	85.4	80.5	87.0	83.9	77.8	79.5	76.7	69.6
(b) MPW-2									
Gas	1.0	2.6	3.4	1.4	2.8	3.9	2.5	3.0	5.1
Liquid	4.8	10.3	10.9	6.2	13.1	15.9	9.0	13.4	21.0
Residue	94.2	87.1	85.3	92.4	84.1	80.2	88.5	83.6	73.9

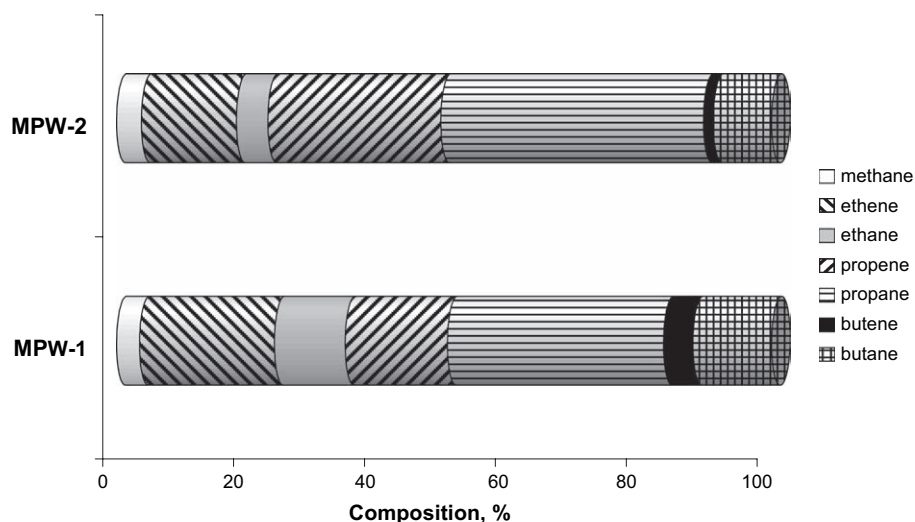


Fig. 3. The composition of gases after cracking of MPW wastes.

significantly different volatile products. Sample MPW-1 consisted of only polyethylene, polypropylene and ethylene-propylene copolymer, therefore the possibility of aromatic content in volatile products was very low. The 0.5–1% aromatic content was a result of the cyclization aromatization reactions of degradation products. It was found that the concentration of aromatics increased with cracking temperature and at 550 °C its concentration was 2.24%. A similar conclusion was found in the experiment of South Korean researchers in batch cracking at 350–450 °C [20]. Simple MPW-2 contained 8% polystyrene, therefore its degradation resulted in liquids with significant aromatic content. Sample MPW-2 contained 8% polystyrene, therefore its degradation yielded liquids with significant aromatic content. The concentration of ethyl-benzene and styrene—the building elements of polystyrene—was evidently dominant, but new aromatics also formed. The concentration of aromatics in the product was significantly higher (20–27%) than the concentration of polystyrene in the raw material (8%). Increasing temperature caused a rise in aromatic content. Further cracking of the polymer chain was observed at higher temperatures, resulting in decreasing concentrations of styrene and ethyl-benzene while the concentration of benzene and toluene increased with temperature.

In case of both samples, it was found that the carbon chain distribution in liquids becomes wider with increasing temperature, e.g. at 550 °C already the C₂₈–C₃₀ hydrocarbons were detectable in the liquids. Less difference was observed in the relation of different compounds, because the concentration of lighter hydrocarbons decreased while the concentration of heavier ones increased with temperature. This can be explained by the heat capacity of melted polymer. At higher cracking temperatures, the temperature of the separator

was also higher, therefore heavier hydrocarbons could be distilled from the melted polymer. This tendency was also detectable by observing the consistency of the liquid phase; with increasing temperature the liquid samples became more and more suspension-like. The residence time had no significant effect on the properties of products, even though increasing residence time fostered the formation of aromatics and greater degradation.

As demonstrated in Table 2, the cracked products (residue and liquid) were further separated in atmospheric and vacuum distillation steps into white spirit-like F1 fraction, diesel fuel-like F2 fraction and a solid residue F3 at room temperature. Both atmospheric and vacuum distillations were carried out in a distillation column filled with Raschig rings. The properties of each fraction are shown in Table 4.

Table 5 demonstrate the yields obtained by distillation of products of cracking samples at different temperatures and residence times. The same tendency was found in both Fig. 4a and Fig. 4b, i.e. with distillation, further liquid fractions (10–25%) could be separated from the residues.

The composition of liquids was analysed by GC; the results are summarized in Table 4. The white spirit-like fraction (F1) contained C₅–C₁₈ hydrocarbons, the main components being C₇–C₉ hydrocarbons in the case of both MPW-1 and MPW-2. Although the concentration of aliphatic C₇–C₉ hydrocarbons was lower in the product obtained from sample MPW-2 than in that obtained from sample MPW-1, the total C₇–C₉ concentration was significantly higher in the case of MPW-2. It was found that mostly ethyl-benzene and styrene constituted the aromatic content. In the F1 fraction obtained from MPW-2 it was 25.29% (*T* = 500 °C, residence time = 0.9 h). The aromatic content was accumulated in the F1 fraction and its concentration in

Table 3
The composition of liquids cracking of MPW-1 and MPW-2

Temperature (°C)		500 °C			525 °C			550 °C		
Residence time (h)		0.6	0.9	1.2	0.6	0.9	1.2	0.6	0.9	1.2
<i>(a) MPW-1</i>										
Aliphatic	C ₅ –C ₆	15.27	17.65	18.70	12.32	15.72	16.36	13.29	13.86	15.99
	C ₇ –C ₉	27.99	29.42	31.90	30.07	28.61	31.92	30.20	29.18	30.12
	C ₁₀ –C ₁₂	21.63	21.18	20.09	19.61	21.31	23.94	18.29	18.22	20.53
	C ₁₃ –C ₁₅	15.27	13.25	17.13	19.17	14.64	16.08	19.46	17.03	15.86
	C ₁₆ –C ₁₈	12.72	11.77	9.45	10.96	8.60	8.06	7.84	7.90	8.60
	C ₁₉ –C ₂₁	4.90	4.71	1.50	3.99	6.14	1.36	2.91	6.26	3.58
	C ₂₂ –C ₂₄	1.71	1.54	0.63	2.73	2.46	0.68	3.12	4.04	2.46
	C ₂₅ –C ₂₇	0.00	0.00	0.00	0.63	1.21	0.00	1.21	1.26	0.62
	C ₂₈ –C ₃₀	0.00	0.00	0.00	0.00	0.00	0.00	1.81	0.23	0.00
	Σ _{aliphatic}	99.49	99.52	99.40	99.48	98.69	98.40	98.13	97.98	97.76
Aromatic	Benzene	0.09	0.12	0.11	0.08	0.16	0.18	0.05	0.09	0.13
	Toluene	0.00	0.00	0.06	0.04	0.06	0.10	0.10	0.18	0.11
	Ethyl-benzene	0.42	0.36	0.43	0.37	1.00	1.20	1.41	1.46	1.57
	Styrene	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	<i>o</i> -Xylene	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	<i>m</i> -Xylene	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	C ₁₆ +	0.00	0.00	0.00	0.03	0.09	0.12	0.31	0.29	0.43
	Σ _{aromatic}	0.51	0.48	0.60	0.52	1.31	1.60	1.87	2.02	2.24
<i>(b) MPW-2</i>										
Aliphatic	C ₅ –C ₆	13.42	14.75	15.86	12.99	13.36	12.96	10.70	9.89	10.81
	C ₇ –C ₉	22.08	22.83	24.79	24.37	23.24	23.03	24.12	22.93	23.78
	C ₁₀ –C ₁₂	18.48	16.95	16.86	15.99	16.45	19.19	14.11	14.83	14.05
	C ₁₃ –C ₁₅	12.65	15.95	15.86	11.99	10.65	13.44	14.59	13.84	13.41
	C ₁₆ –C ₁₈	10.41	7.97	6.94	8.88	6.78	6.72	6.81	6.92	7.57
	C ₁₉ –C ₂₁	3.70	2.99	1.98	3.50	4.84	1.92	2.63	3.95	3.24
	C ₂₂ –C ₂₄	0.97	1.38	0.40	2.30	1.94	2.40	2.92	4.15	2.16
	C ₂₅ –C ₂₇	0.00	0.00	0.00	0.60	0.95	0.00	0.97	0.96	0.54
	C ₂₈ –C ₃₀	0.00	0.00	0.00	0.00	0.00	0.00	1.65	0.18	0.00
	Σ _{aliphatic}	81.71	82.82	82.69	80.61	78.21	79.65	78.50	77.65	75.57
Aromatic	Benzene	0.08	0.07	0.18	0.99	0.65	0.80	0.91	1.31	1.37
	Toluene	0.49	0.67	0.69	1.60	2.03	1.92	2.24	2.96	3.24
	Ethyl-benzene	0.70	0.70	0.61	0.81	1.00	1.23	1.69	1.85	2.31
	Styrene	10.70	9.77	9.91	10.59	11.33	9.00	10.70	10.18	9.84
	<i>o</i> -Xylene	1.65	1.00	1.27	1.30	1.74	1.14	1.36	1.94	2.59
	<i>m</i> -Xylene	0.39	0.30	0.29	0.28	0.77	1.15	0.39	0.65	0.86
	C ₁₆ +	4.28	4.68	4.34	3.82	4.26	3.9	4.18	3.46	4.22
	Σ _{aromatic}	18.29	17.18	17.31	19.39	21.79	20.35	21.50	22.35	24.43

F2 was below 1%. This is an advantageous property for further, fuel-like or petrochemical applications, because the aromatic content in white spirit does not cause problems; the low aromatic and high branched hydrocarbon content in diesel gas oil is also an advantageous property.

The olefin content of liquids and their distribution are of importance in further applications. Fig. 5a–d shows the infrared spectra of liquids in the 400–4000 cm^{−1} wavenumber range. Significant differences were observed in the samples investigated. The stretching vibrations in the 800–1000 cm^{−1} wavenumber range gave information about different olefin double bonds. The vinyl type double bonds gave two intensive adsorption bands at 910 and 990 cm^{−1}, the vinylidene at wavenumber 890 cm^{−1} and the internal double bonds at 956 cm^{−1}. In the case of both MPW-1 and MPW-2 samples, it was found that all three types of double

bonds were present in every fraction; there were more internal double bonds in fractions F1 than in fractions F2. Another result was found in the case of aromatic compounds formed mostly in the cracking of polystyrene. It was demonstrated earlier that the degradation of polystyrene in a batch reactor already degraded at 360–370 °C into volatile products. They consist of mostly benzene, toluene, ethyl-benzene, styrene, and dimer, trimer, and tetramer of the monomer. It was described that the probability of formation of heavier fragments than tetramer was low [18]. The sample of MPW-2 had 8% polystyrene content, but as demonstrated, the concentration of aromatic compounds was 20–30%, concentrated in fraction F1.

The CH stretching vibrations band between 1600 and 1650 cm^{−1}—observable in Fig. 5a—gave a split band, because different substituents are attached to the aromatic ring. The intensity and the location of the split

Table 4

The properties of fractions after distillation (temperature 520–530 °C, residence time 0.9 h)

Fraction	MPW-1		MPW-2	
	F1	F2	F1	F2
Density g cm ⁻³	0.755	0.803	0.749	0.801
<i>Composition</i>				
<i>Aliphatic</i>				
C ₅ –C ₆	23.07	0.00	15.23	0.0
C ₇ –C ₉	40.30	0.00	34.05	0.0
C ₁₀ –C ₁₂	30.04	10.43	21.32	7.36
C ₁₃ –C ₁₅	5.66	19.12	3.93	22.77
C ₁₆ –C ₁₈	0.14	26.19	0.18	22.60
C ₁₉ –C ₂₁	0.00	22.83	0.0	22.60
C ₂₂ –C ₂₄	0.00	17.84	0.0	22.60
C ₂₅ –C ₂₇	0.00	3.59	0.0	1.49
Aromatic	0.78	0.0	25.29	0.58
<i>Groups in average molecule</i>				
–CH ₂ –	8.9	16.8	8.0	15.5
–CH ₃	3.1	1.7	4.2	1.9
<i>Olefin content</i>				
Vinyl	18.4	13.9	16.8	12.3
Vinylidene	22.7	15.2	18.1	14.4
Internal	3.2	4.6	2.7	4.2
VK ₄₀ (mm ² s ⁻¹)	–	3.6	–	3.3
Pour point (°C)	–25	–4	–23	–4
Flash point (°C)	–	83	–	101
Corrosion test	Group 1	Group 1	Group 1	Group 1
S content (ppm)	3	4	52	5
N content (ppm)	29	31	1459	1142
Boiling range (°C)	39–195	179–303	35–204	189–307
M (g mol ⁻¹)	108	210	104	213

bands was a function of the type of substitution and their electro-negativity. The degenerated vibration pair at 1610 cm⁻¹ in the case of toluene between 1610 and 1580 cm⁻¹ wavenumber appears with medium intensity. Another band also referring to aromatics was found between 800 and 700 cm⁻¹.

The heteroatom content of liquids was determined according to ASTM-D 6428 (sulphur) and ASTM-6366 (nitrogen). The concentration of both sulphur and nitrogen is an important parameter for further fuel-like application of liquids, because their low heteroatom content is an advantage. It was supposed that polymers containing

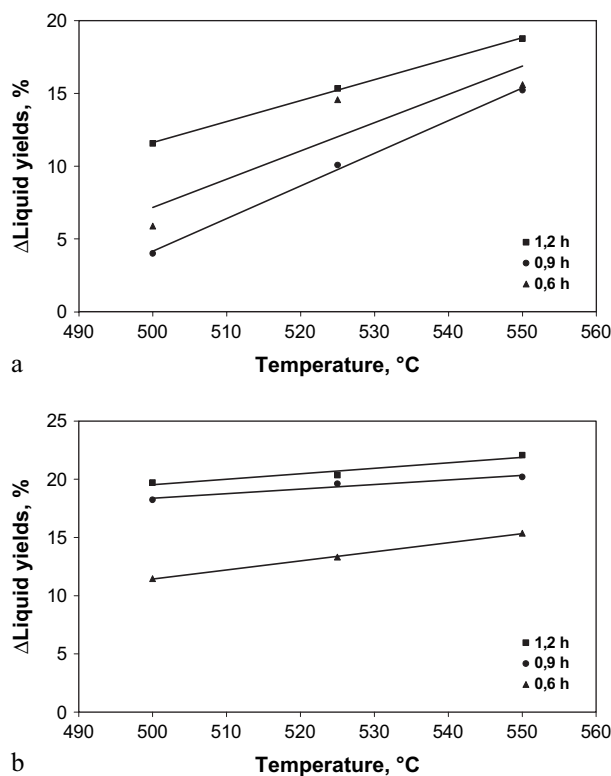


Fig. 4. (a) The difference in liquid yields obtained without distillation and by distillation (MPW-1). (b) The difference in liquid yields obtained without distillation and by distillation (MPW-2).

additives with heteroatoms (flame-retardant, antismoke additive) were degraded, and the volatile decomposition products with sulphur and nitrogen content condensed in the fraction of heavier hydrocarbons. Experimental results are summarized in Fig. 6a,b. The concentration of sulphur was under 10 ppm in the case of liquids produced from MPW-1, while 50–80 ppm of sulphur content was found in the other sample (MPW-2). The sulphur content of the F1 fraction from the MPW-2 sample had considerable higher sulphur content than F2, because the sulphur-containing compounds had lower boiling points. No tendency was observed concerning cracking parameters and sulphur content.

Table 5

The yields after distillation of MPW-1 and MPW-2 products

Temperature (°C)	500 °C			525 °C			550 °C		
Residence time (h)	0.6	0.9	1.2	0.6	0.9	1.2	0.6	0.9	1.2
<i>(a) MPW-1</i>									
Gas	1.0	2.3	4.1	1.9	2.7	5.0	3.0	3.9	6.1
F1	8.0	11.0	13.8	9.0	12.4	14.3	12.3	16.5	17.4
F2	10.0	19.5	21.3	15.4	20.7	23.3	20.5	23.1	28.9
F3	81.0	67.2	60.8	78.2	64.5	57.5	68.6	53.6	47.5
<i>(b) MPW-2</i>									
Gas	1.0	2.6	3.5	1.4	2.8	3.9	2.5	3.0	5.1
F1	3.8	6.4	9.1	10.5	11.2	14.7	11.5	13.5	20.2
F2	6.9	7.9	13.4	10.2	12.0	16.6	13.1	15.1	19.5
F3	88.3	83.1	74.2	77.9	74.0	64.9	72.9	68.4	55.2

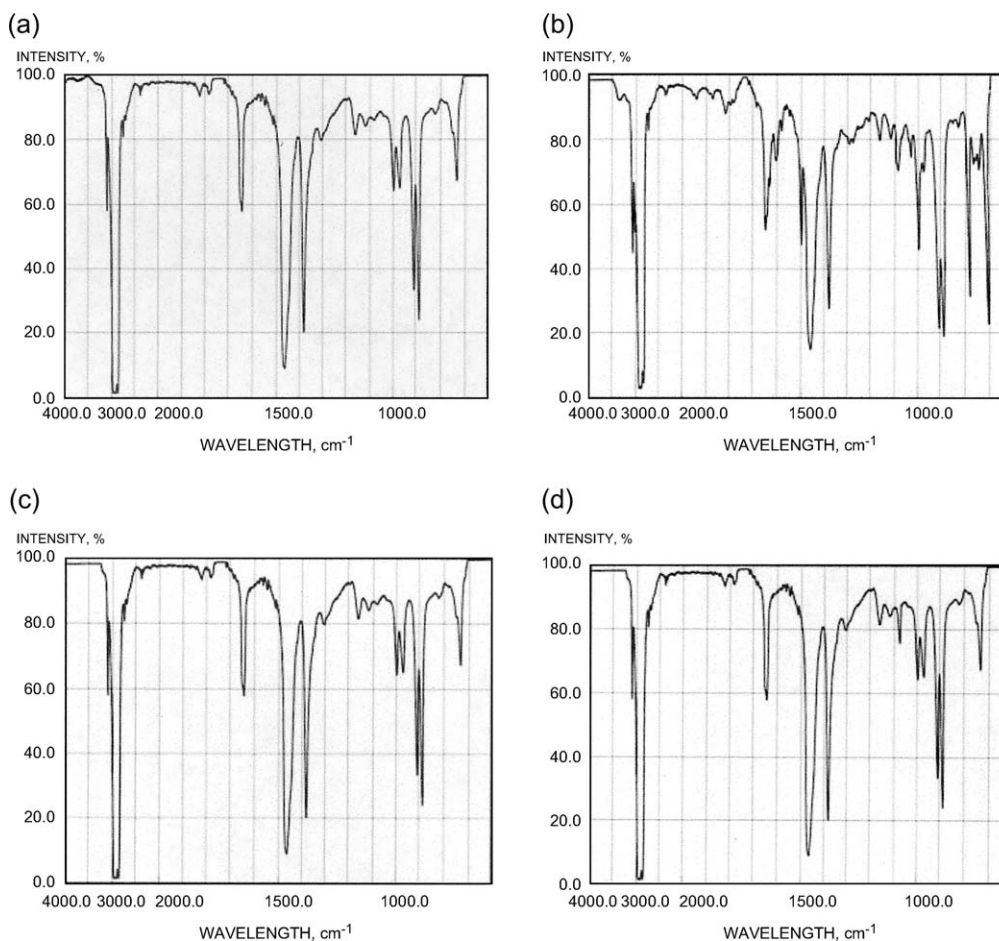


Fig. 5. (a) The infrared spectrum of liquid (MPW-1, F1). (b) The infrared spectrum of liquid (MPW-2, F1). (c) The infrared spectrum of liquid (MPW-1, F2). (d) The infrared spectrum of liquid (MPW-2, F2).

The nitrogen content of liquids obtained by distillation is demonstrated in Fig. 6b. Liquids from sample MPW-1 had very low (under 20 ppm) nitrogen content. In the cracking of polyamide containing MPW-2, it was found that the nitrogen concentration in liquids increased by three orders of magnitude. The nitrogen appeared not only in one fraction but in both fractions (F1 and F2). This means that the nitrogen containing compounds formed had a wide boiling point range. Increasing nitrogen content was observed with decreasing residence time and temperature. In the cracking of MPW-2, each case resulted in higher nitrogen content in the lighter F1 fraction than in the F2 fraction.

The main properties of the heavy oil-like F3 fraction obtained from the residue are given in Table 6. Both composition of waste samples and cracking temperature significantly affected the structure of the hydrocarbons formed. As a consequence of a greater ratio of cracking of C–C bonds, the density of residue decreased with cracking temperatures. The distribution of olefin double bonds were analysed by the infrared method, as demonstrated above. Mainly α -olefins (vinyl and vinylidene)

constitute the gross olefin content. Similar results are shown in Table 4. The concentration of olefins of F3 was smaller than in case of F1 and F2. Flash and pour points of heavy oil fractions obtained from MPW-1 waste samples were lower than from MPW-2. This could be explained by the chemical structure. F3 fractions from the cracking of MPW-1 had more branched hydrocarbons, because MPW-1 contains a higher ratio of polypropylene than MPW-2. The concentration of metals from pigments and filler materials was measured by EDXRF spectrometer. It was found that the concentration of metals in the heavy oil-like residue was higher than in the raw materials as a consequence of the removal of lighter products (F1 and F2).

4. Conclusions

The thermal degradation of different municipal plastic wastes containing polyethylene, polypropylene, ethylene-propylene copolymer, polystyrene, polyamide and polyurethane rubber was investigated in a horizontal

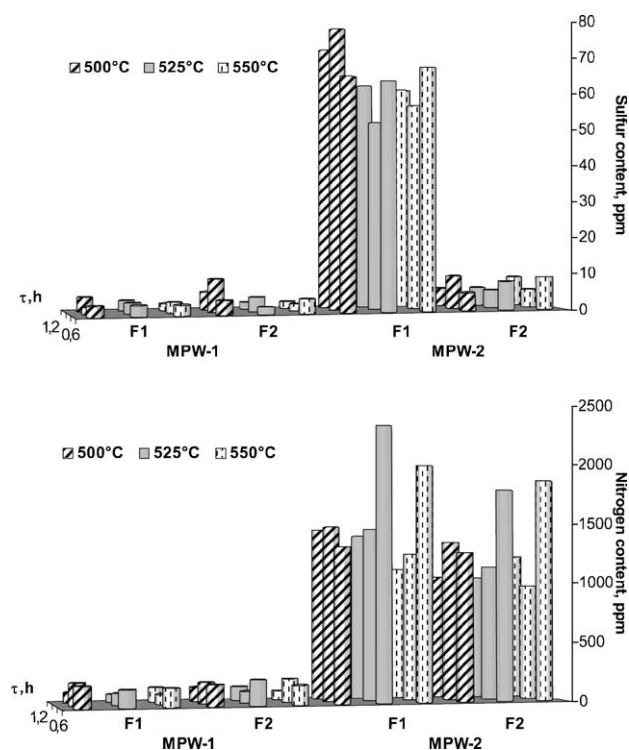


Fig. 6. (a) The sulphur content of liquids. (b) The nitrogen content of liquids.

tube reactor using different cracking parameters. The further utilization of volatile products, i.e. fuel-like utilization was investigated; moreover, the effects of the cracking parameters (temperature, residence time) on the yields and properties of products were determined.

It was found that the chemical structure of polymers greatly affected the qualitative and quantitative properties of volatile products. The yields of volatile products increased both with temperature and residence time. The yields of liquids of 10–25% separated from the gases leaving the reactor could be increased further by 10–15% by distilling the residue. When polystyrene

was also in the polymer blend, a significant aromatic content was observed, mainly ethyl-benzene, styrene, toluene and benzene. After distillation, the aromatic content was concentrated mostly in the lighter fractions (F1), which is an advantageous property for further fuel-like utilization. Similar to the aromatic content, the heteroatom content of product liquids was also influenced by the structure of the raw material. Cracking the MPW-2 sample yielded a liquid with high sulphur and nitrogen content. This is a disadvantageous property for further utilization. It was found that the liquids had approximately 50% olefin content depending on cracking parameters.

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Table 6
The properties of heavy oil-like fractions after distillation

Sample	MPW-1			MPW-2		
Temperature (°C)	500	525	550	500	525	550
Density (g cm ⁻³)	0.853	0.849	0.838	0.859	0.852	0.840
Olefin content (%)						
Vinyl	11.3	12.1	11.8	9.8	9.4	10.0
Vinylidene	14.7	14.9	14.4	11.4	10.6	11.3
Internal	2.5	2.3	2.6	2.3	2.5	2.5
I/Br number (g I/100 g product)	41	43	42	41	42	40
Pour point (°C)	158	145	130	164	151	136
Flash point (°C)	272	258	239	290	263	244
Metals (ppm)						
Ca	511	549	672	283	305	368
Zn	409	468	537	187	216	254
Ti	688	762	875	363	428	456

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